INFO-F420 COMPUTATIONAL GEOMETRY · ULB · 2026–2027

How we shared the work

Guarding Walls Contiguously — Ilias Akajou (520466) and Jomaa Abdulrahman (583629)

> **DRAFT — to be checked and completed by both of us so that it describes what each of us actually did.**

1. Change with respect to the proposal

In the proposal, Ilias was responsible for part 1 (the problem) and Jomaa for part 2 (the algorithm). Following the feedback we received ("do not spend too much time on classical art gallery problems; both work on part 2 and split the explanations there; use web widgets to illustrate the different steps of the proof"), we reduced part 1 to a short introduction and split the explanations of part 2 between us.

2. Explanations of the web page

Section	Topic	Paper	Responsible
1	The problem, context (short)	§1–2	Ilias
2.1	Visibility cores, one-guard case	Obs. 2.1, 2.2	Ilias
2.2	Greedy algorithm, k + 1 bound	Thm 7.1 (proof)	Ilias
2.3	Good and bad guards	Def. 4.1, Obs. 4.8	Ilias
2.4	Replacing guards	Prop. 4.3, Lemmas 4.2–4.6, Cor. 4.7, Lemma 4.9	Ilias
2.5	Candidate guards and start points	Def. 5.1, 5.2, Lemma 6.6	Jomaa
2.6	Theorem 5.4	Lemma 5.3, Thm 5.4	Jomaa
2.7	Maximal visible chains	Def. 6.1, Lemmas 6.2, 6.5	Jomaa
2.8	One greedy step in $O(\log^2 n)$	Lemma 6.8	Jomaa
2.9	Complete algorithm, running time	Thm 7.1	Ilias
2.10	The $\Theta(n \log n)$ algorithm	[6]	Jomaa
3	Game and sandbox	—	both
4	Summary, implementation, references	—	both

Each subsection of the web page shows its author ("Explained by …").

3. Implementation

Component	Files	Responsible
Geometry kernel (visibility, cores, shortest paths)	src/geometry/vec, polygon, visibility, core, shortestPath	to complete
Solver (G_P, U_P, next, greedy)	src/geometry/arrangement, solver	to complete
Proof constructions and their figures	src/geometry/proofs, src/widgets (Figures 2–11)	split as in section 2
Game, sandbox, random polygons	src/widgets/GameWidget, src/geometry/generate	to complete
Tests	tests/	to complete
Page layout, report text	src/App, src/sections	both

4. Use of AI tools

We used an AI assistant during the project; how and for what is described in our AI journal, which is handed in with this document.